

Example 1. Let n be an integer greater than 1.

We say n is **square-free** iff n **not** divisible by any perfect square bigger than 1.

Prove n is square-free if and only if it is:

$n =$

Solution. $\implies$: Assume n is square-free.

Then its prime decomposition cannot contain any factor:

because:

Hence its prime decomposition is:

$\impliedby$: Assume n is:

Suppose towards a contradiction that it were divisible by

Select a prime factor:

Then p^2 also:

Thus the prime decomposition of n contains p^k for:

But the prime decomposition is unique.

This contradicts that there is a prime decomposition of n as

□

A. **Sieve of Eratosthenes.** There is slick method for enumerating primes.

Its theoretical foundation is the following.

Theorem. Let n be an integer greater than 1.

If n is composite, then it has a nontrivial positive divisor:

Nontrivial means neither ± 1 nor $\pm n$.

Proof. Let n be composite. Select a nontrivial factorization:

$n =$

It cannot be the case that both a and b are greater than $\sqrt{n}$ for otherwise:

Hence one of a and b is a positive nontrivial divisor less or equal $\sqrt{n}$.

□

So to check whether an integer is prime, one need only test for nontrivial divisors

From this comes the technique, the **Sieve of Eratosthenes**, for enumerating primes.

First list out 2 through n .

Next cross out all multiples of:

The survivors are the primes from 1 to n .

In the table below we cross out all multiples of all primes:

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Example 2. The repunit R_n is defined by

$$R_n = \underbrace{11 \dots 1}_{n \text{ ones}}$$

Find the prime factorization of:

$$R_4 = 1111$$

using the following, which was established in the MyOpenMath A2 Practice.

- If d is a positive integer such that $d \mid n$, then:

Solution. Since $2 \mid 4$ it follows that:

We execute long division.

For the remaining factor, we check for divisibility by primes less than:

Or in other words we check divisibility by primes:

So the prime decomposition is

$$1111 =$$

B. Divisors. We can use prime decompositions to describe divisors.

Theorem. Let n be an integer greater than 1 with prime decomposition:

$$n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

Then the positive divisors of n are precisely the integers d of the form:

$$d =$$

where natural numbers:

For example we list the divisors of $12 =$

Example 3. Let n be a positive integer and p be prime so $n < p < 2n$.

Prove that p divides

$$\binom{2n}{n}$$

Solution. All binomial coefficients are:

By definition we may write as a fraction of integers

$$\binom{2n}{n} =$$

In terms of prime decompositions, division producing an integer is executed by

Because $p < 2n$ it follows that

Because p is prime and greater than n it follows that

as otherwise p would have to divide:

which is not possible.

So when division by cancelling prime factors is done, the resulting integer has

Hence the resulting integer is divisible by p .

□